

bKash PGW

Tokenized Checkout Integration

API Documentation

Sheikh Pharma | sheikhpharma.shop

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1. Overview

This document describes the bKash PGW (Payment Gateway) Tokenized Checkout V2 integration for Sheikh Pharma (sheikhpharma.shop). All APIs are implemented as Supabase Edge Functions with 30-second timeout.

Base URL: <https://vkfloynvxqkxbqpdzbd.supabase.co/functions/v1>

Callback URL: <https://sheikhpharma.shop/bkash/callback>

Demo URL: <https://merchantdemo.sandbox.bka.sh/tokenized-checkout/version/v2>

Sandbox Base: <https://tokenized.sandbox.bka.sh/v2>

Production Base: <https://tokenized.pay.bka.sh/v2>

⚠ API timeout of 30 seconds is set for ALL APIs. Site must have valid SSL certificate (TLS 1.2 or higher).

2. Required APIs Summary

#	API Name	Endpoint	Note
1	Grant Token	(Internal - auto-cached 50min)	Required
2	Create Payment	/bkash-create-payment	Required
3	Execute Payment	/bkash-execute-payment	Required
4	Query Payment	/bkash-execute-payment (auto-fallback)	Mandatory if Execute timeout
5	Search Transaction	/bkash-search-transaction	Required
6	Refund	/bkash-refund (action: refund)	If applicable
7	Refund Status	/bkash-refund (action: refund_status)	Mandatory if Refund timeout

3. Grant Token API

Automatically called internally before every bKash API call. Tokens are cached for 50 minutes to reduce API calls.

bKash Endpoint

```
POST {bkash_base_url}/tokenized-checkout/auth/grant-token
```

Request Headers

Parameter	Type	Description
Content-Type	string	application/json
Accept	string	application/json
username	string	bKash merchant username
password	string	bKash merchant password

Request Body

```
{ "app_key": "...", "app_secret": "..." }
```

Response

```
{ "id_token": "eyJhbGciOiJIUzI1...", "token_type": "Bearer", "expires_in": 3600 }
```

⚠ Token is cached for 50 minutes. Auto-refreshed when expired.

4. Create Payment API

Initializes a bKash payment and returns a redirect URL for the customer to complete payment.

Endpoint

POST <https://vkfloynvxqkxbqpdIzbd.supabase.co/functions/v1/bkash-create-payment>

Request Body

Parameter	Type	Description
amount	number	Payment amount in BDT (required)
order_id	string	Order UUID from the system (required)
callback_url	string	URL to redirect after payment (required)

Example Request

```
{ "amount": 500, "order_id": "abc-123-def", "callback_url":  
  "https://sheikhpharma.shop/bkash/callback" }
```

Success Response (200)

```
{ "bkashURL": "https://tokenized.pay.bka.sh/...", "paymentID": "TR001234" }
```

Error Response (400)

```
{ "error": "Create payment failed", "errorCode": "2001" }
```

⚠ 30-second timeout applied.

5. Execute Payment API

Called after the customer completes payment on bKash. Confirms the transaction server-side.

Endpoint

POST `https://vkfloynvxqkxbqpdzbd.supabase.co/functions/v1/bkash-execute-payment`

Request Body

Parameter	Type	Description
paymentID	string	Payment ID from bKash callback (required)
order_id	string	Order UUID from the system (required)
payment_type	string	Optional: 'cod_delivery_charge', 'lab_service', or empty for full payment

Success Response (200)

```
{ "success": true, "trxID": "9HK3001F5E", "amount": "500" }
```

Error Response (400)

```
{ "success": false, "error": "Payment not completed", "statusCode": "2023" }
```

⚠ If Execute does not respond within 30 seconds, Query Payment API is automatically called as fallback.

6. Query Payment API (Mandatory Fallback)

If Execute Payment does not return a response within 30 seconds, this API is automatically called to check payment status. This is MANDATORY as per bKash requirements.

bKash Endpoint (called internally)

POST `{bkash_base_url}/tokenized-checkout/payment/status`

Request Body

```
{ "paymentID": "TR001234" }
```

Response

```
{ "paymentID": "TR001234", "trxID": "9HK3001F5E", "transactionStatus": "Completed", "amount": "500" }
```

⚠ Built into Execute Payment function. No separate endpoint needed.

7. Search Transaction API

Search for any transaction using the bKash Transaction ID (trxID).

Endpoint

POST <https://vkfloynvxqkxbqpdIzbd.supabase.co/functions/v1/bkash-search-transaction>

Request Body

Parameter	Type	Description
trxID	string	bKash Transaction ID (required)

Example Request

```
{ "trxID": "9HK3001F5E" }
```

Response

```
{ "trxID": "9HK3001F5E", "initiationTime": "2026-04-13T10:00:00",  
  "completedTime": "2026-04-13T10:01:00", "transactionStatus": "Completed",  
  "amount": "500", "currency": "BDT", "customerMsisdn": "01XXXXXXXXXX" }
```

8. Refund API

Process a refund for a completed bKash transaction. If refund response is not received within 30 seconds, Refund Status API is automatically called as fallback.

Endpoint

POST <https://vkfloynvxqkxbqpdldbd.supabase.co/functions/v1/bkash-refund>

Request Body

Parameter	Type	Description
action	string	"refund" (required)
paymentID	string	Original bKash Payment ID (required)
trxID	string	Original bKash Transaction ID (required)
amount	number	Refund amount in BDT (required)
reason	string	Reason for refund (optional)
sku	string	Product SKU (optional)

Example Request

```
{ "action": "refund", "paymentID": "TR001234", "trxID": "9HK3001F5E",  
  "amount": 500, "reason": "Customer return" }
```

Success Response

```
{ "originalTrxID": "9HK3001F5E", "refundTrxID": "9HK3001F5F",  
  "transactionStatus": "Completed", "amount": "500" }
```

9. Refund Status API (Mandatory)

Check the status of a refund. MANDATORY if refund response is not received within 30 seconds, or to check refund status anytime.

Endpoint

POST <https://vkfloynvxqkxbqpdldbd.supabase.co/functions/v1/bkash-refund>

Request Body

Parameter	Type	Description
action	string	"refund_status" (required)
paymentID	string	Original bKash Payment ID (required)
trxID	string	Original bKash Transaction ID (required)

Response

```
{ "originalTrxID": "9HK3001F5E", "refundTrxID": "9HK3001F5F",  
  "transactionStatus": "Completed", "amount": "500" }
```

10. Payment Flow (User Journey)

1. Customer adds products to cart and proceeds to checkout
2. Customer selects bKash as payment method
3. System calls Create Payment API to get bKashURL
4. Customer is redirected to bKash payment page
5. Customer enters wallet number, receives OTP, enters PIN
6. bKash redirects to callback URL: <https://sheikhpharma.shop/bkash/callback>
7. Callback page calls Execute Payment API with paymentID
8. If Execute times out (30s), Query Payment API is called automatically
9. On success: order is marked as paid, customer sees confirmation
10. On failure: customer is shown error with retry option

11. Sandbox Test Credentials

Use these wallet numbers for testing in sandbox environment:

Test Wallet Numbers

Wallet Number	PIN
01770618575	12121
01929918378	12121
01770618576	21212
01877722345	13131
01619777282	21212
01619777283	13131

OTP for all wallets: 123456

⚠ If one wallet gets locked/inactive, use another one. PIN and OTP are same for all wallets.

12. Integration Milestones

Stage	Milestone	Tasks
S-1	Integration Initiation	Developer Portal, Demo Link, Sandbox-1 credentials received
S-2	System Readiness (Sandbox)	a. Confirm dev readiness b. Share user journey flow c. Solution document finalization
S-3	Sandbox#2 Validation	a. bKash shares Sandbox-2 b. Share required API responses c. bKash validates (2 days)
S-4	Production Info	Share MSISDN, Display Name, etc.
S-5	Production Onboarding	bKash provisions on Production (1-2 days)
S-6	Production Readiness	Confirm system readiness with Production credentials
S-7	UAT & Go Live	a. Business UAT b. Technical UAT (2-3 days) c. Payment live for all customers

13. Credentials Storage

All bKash credentials are securely stored as Supabase Edge Function secrets and also configurable via admin panel (site_settings):

Parameter	Type	Description
BKASH_APP_KEY	string	bKash App Key
BKASH_APP_SECRET	string	bKash App Secret
BKASH_USERNAME	string	bKash Merchant Username
BKASH_PASSWORD	string	bKash Merchant Password
BKASH_BASE_URL	string	bKash API Base URL (sandbox or production)

14. Error Handling

All error codes follow bKash PGW standard error codes:

Reference: <https://developer.bka.sh/docs/error-codes>

⚠ 30-second timeout is enforced on all API calls. If timeout occurs on Execute, Query Payment is called. If timeout occurs on Refund, Refund Status is called.

15. Security Requirements

- Valid SSL certificate required on merchant site
- TLS 1.2 or higher supported only
- All credentials stored server-side (Edge Functions)
- No credentials exposed to client/browser
- Payment verification done server-side before order confirmation